Aerospace Manufacturing Core
Study Guide

"AMCSG-201"

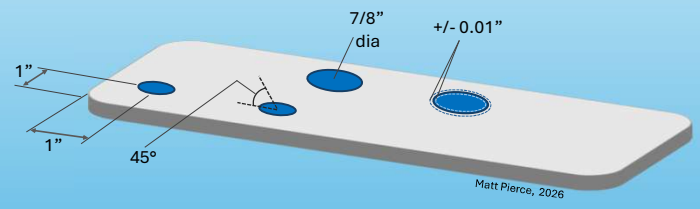
Tools & Processes

Drilling & Countersinking

Drilling

Drilling

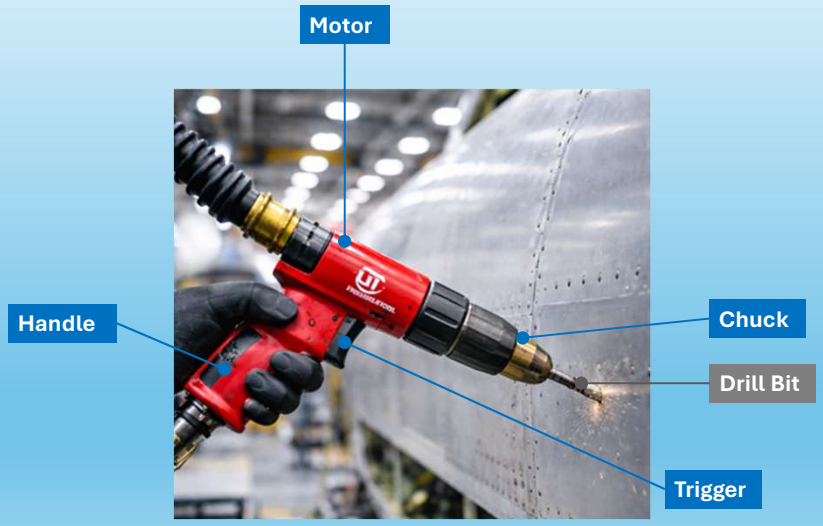
- Drilling a quality hole is crucial
- Used to fasten parts together using rivets, screws, bolts, etc.
- A **quality hole** must meet requirements for:
  - **Location**
  - **Angularity**
  - **Dimension**
  - **Tolerance**
  - Free of burrs & sharp edges



Public Domain Image Sources:

- I drew the diagram in PPT

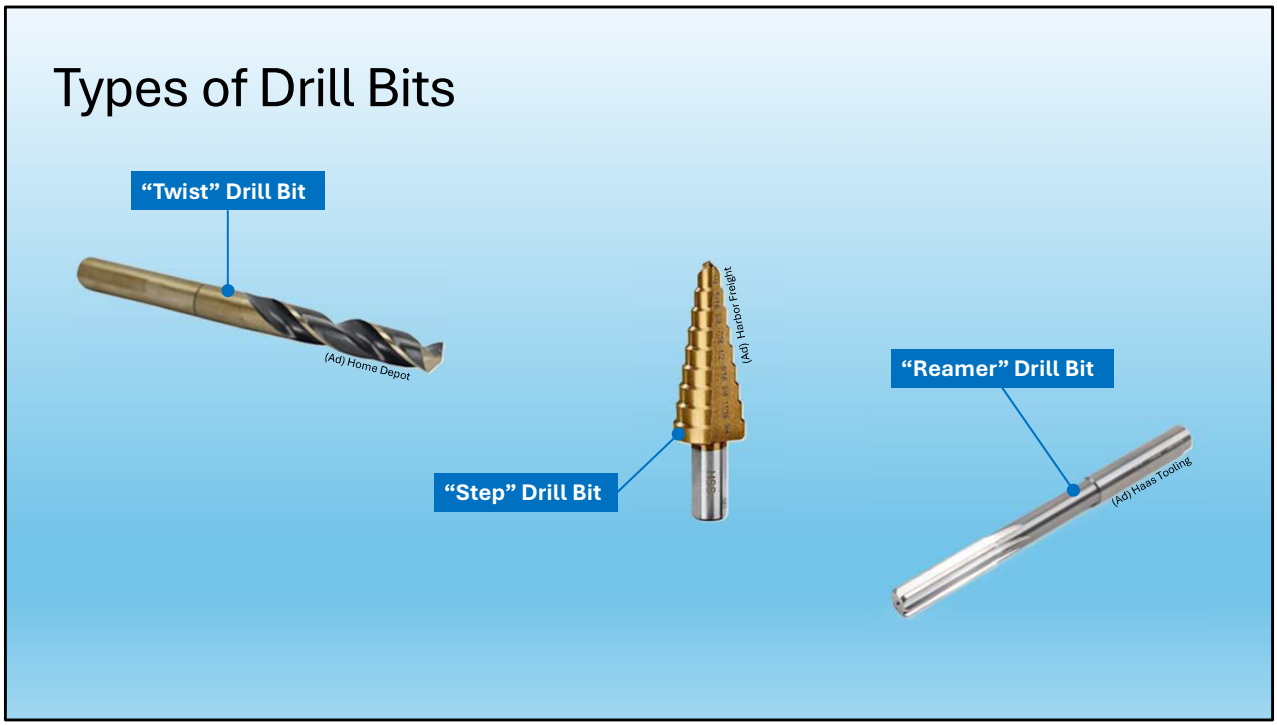
Basic Components of Drill Motor



ChatGPT / Matt Pierce, 2026

Image Credits:

- ChatGPT "create a picture of a drill motor used in aerospace manufacturing" + "make it look like a universal tool drill motor in red color with a handle"



Twist Drill Bit = Make a hole
Step Drill Bit = Make holes at multiple diameters without switching bits
Reamer Bit = Make a hole larger with precision

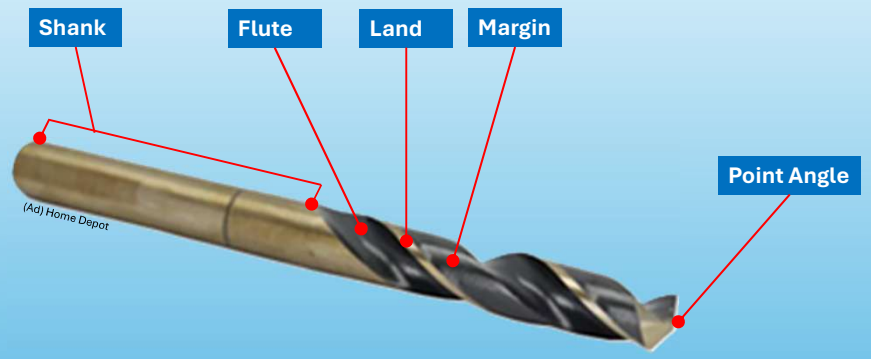
<u>Image Credits</u>:

- Twist Drill Bit – Home Depot for sale – (Ad) https://encrypted-tbn0.gstatic.com/shopping?q=tbn:ANd9GcQeeoYPcu5ci_x4HR2VdH7y6-sAVXKNv0LOCUGrIO9LadYaQLuCxAW8ZxF8_KiZZili1DXalQfKcaFhSHnGe4QnUZygBEqkz3k3qub5YYnc_PZSeOIK9Xb7Ew
- Step Drill Bit – Harbor Freight for sale - (Ad) https://www.google.com/aclk?sa=L&ai=DChsSEwigvfGRjLySAxUCAK0GHbEyIF4YACICCAEQCxoCcHY&co=1&ase=2&gclid=CjwKCAiAs4HMBhBJEiwACrfNZfmzqSee-4-C3zrzit-f7FbVp14o5LOHpKcNJWzOQ_cXnjtj3BtAshoCmlcQAvD_BwE&cid=CAASZeRoMeva1YcWMovAG3zCu1ZF-pUUXTMIbAz6ylgte3YUbEInkB6XPFENRREicgoWGOyb2CSjRDL0GUtQRDJuMlX0ghYFg7PfBKyvW5NW-Zm-rJnSdBWfSNibxp5D9M1d3pNzDTzg&cce=2&category=acrcp_v1_32&sig=AOD64_2GqZQNW_kK2rRhwE5bHvL6ssmgbA&ctype=46&q=&ved=2ahUKEwjXsOuR

jLySAxUdMzQIHbffBJMQ9aACKAB6BAgFEEs&adurl=

- Reamer Drill Bit = Haas Tooling website (for sale) = https://www.haastooling.com/content/dam/haas-tooling/ecommerce/products/03/3239/image/03-3239.jpg/_jcr_content/renditions/original./03-3239.jpg

Parts of a Drill Bit



- Higher Point Angle = harder materials (metal)
- Lower Point Angle = softer materials

Public Domain Image Sources:

- Twist Drill Bit – Home Depot for sale - https://encrypted-tbn0.gstatic.com/shopping?q=tbn:ANd9GcQeeoYPcu5ci_x4HR2VdH7y6-sAVXKNv0LOCUGrIO9LadYaQLuCxAW8ZxF8_KiZZili1DXalQfKcaFhSHnGe4QnUZygBEqkz3k3qub5YYnc_PZSeOIK9Xb7Ew

Drill Guides and Drill Stops

- **Drill Guide / Drill Block** to control angle (e.g.: force perpendicular)
- **Drill Stop** to control depth (e.g.: do not puncture below)

Drill Guide (Drill Block not shown)

ChatGPT / Matt Pierce, 2026

Drill Stop

ChatGPT / Matt Pierce, 2026

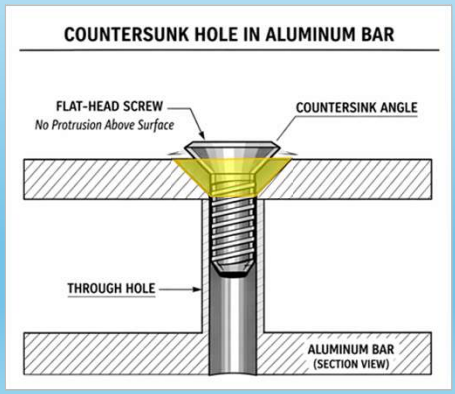
<u>Public Domain Image Sources</u>:

- Drill Guide: ChatGPT "make an image of a red pneumatic "universal tool" drill motor with handle drilling perpendicular into aircraft skin using a perpendicular drill guide"
  + "Use a clear round plastic drill guide, not the black L metal guide"
- Drill Stop: Same ChatGPT above, but modified… "Change the drill guide into a drill stop. show the locking collar and spring. show a side view with drill penetrating the aluminum skin only so far"

Countersinking

Countersinking

- **What**: Countersink creates a conical hole so fastener sits flush
- **Why**: Flush heads improve aerodynamics, reduce corrosion, etc.



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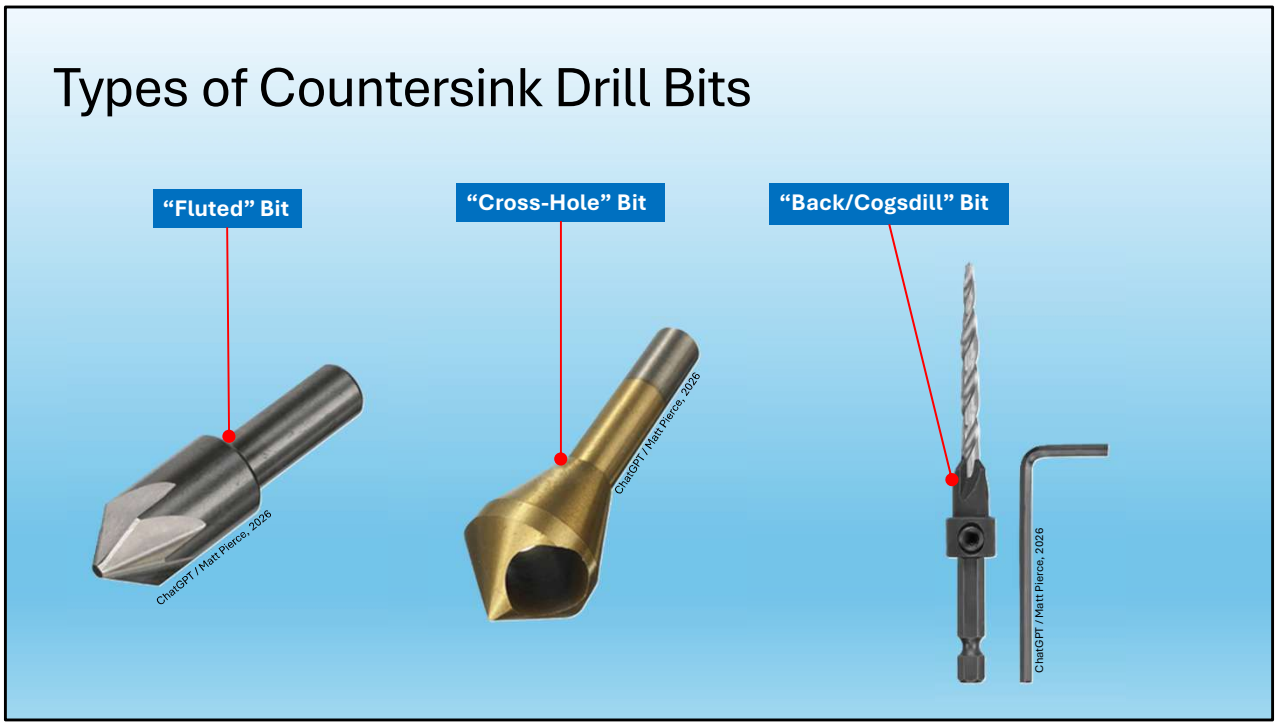
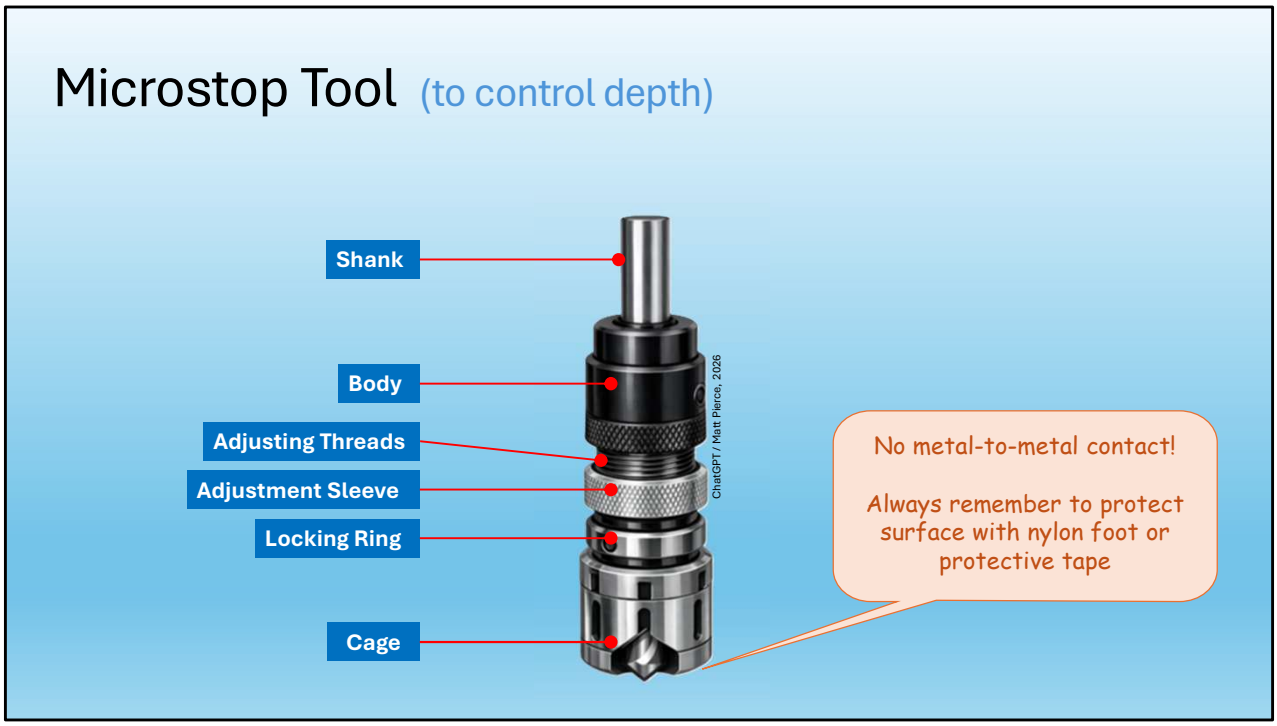
Public Domain Image Sources:

- Countersink hole image: ChatGPT "Make a diagram of a countersink hole thru an aluminum bar indicating how a screw head would sit flush"
  + "Redraw this as an ASTM-style engineering diagram. Format it exactly for a PowerPoint or training manual. One diagram. No drill stops. No drill guides."



- Three images from ChatGPT: "show me a picture of three different countersink bits: (1) Fluted countersink bit, (2) Cross-Hole countersink bit, and (3) Back countersink bit (aka Cogsdill)."
  + "Make all pictures show just one bit each, standing perpendicular"



- Image from ChatGPT: “make a diagram showing a microstop tool (for countersinking). Be sure to show the shank, the body, the adjusting threads, the adjustment sleeve, the locking ring, and the cage. Do not label anything.”

Q&A

Practice Q&A's (Q1-Q8)

Q1: What four requirements make a quality drilled hole?

A: Location, Angularity, Dimension, and Tolerance

Q2: What is the most common type of drill bit?

A: Twist drillbit

Q3: What type of drillbit lets you drill increasingly larger diameter holes without changing bits?

A: Step drillbit

Q4: What part of the drillbit is clamped into the chuck?

A: Shank

Q5: What purpose does the flute part of a drillbit server?

A: Allows waste material to exit the hole

Q6: What tool do you use when drilling to force perpendicular holes?

A: Drill guide

Q7: What tool do you use when drilling to control depth of the hole?

A: Drill stop

Q8: What is a hole called with a conical top into which a flat-head fastener can sit flush?

A: Countersink

Practice Q&A's (Q9-Q16)

Q9: Name three types of countersink drillbits?

A: 1. Fluted bit, 2. Cross-Hole bit, 3. Cogsdill bit

Q10: What tool do you use when drilling a countersink hole to control depth?

A: Microstop tool

Q11: What is not allowed for drillbits over six inches long?

A: Drilling speeds > 6,000RPM

Q12: Why is it important to drill quality holes?

A: Lowers cost and re-work. Increases airframe lifespan

Q13: Where is the diameter of the drillbit stamped?

A: On the shank

Q14: How is a chamfered hole different from a countersink hole?

A1: Chamfer is 45°, shallow, for safety/deburring, and easy fastener entry.

A2: Countersink is 82°-90°, deeper, tighter tolerance

Q15: What type of surface is most difficult to countersink?

A: Curved

Q16: How frequently should you test countersink holes?

A: Every 10 to 20 holes

Practice Q&A's (Q17-Q19)

Q17: What is the name of the extra material in a slight radius between the shank and the head of a fastener used to strengthen it to avoid snapping there when you turn with screwdriver or wrench?

A: Fillet

Q18: What keeps the Cogsdill bit accurately centered in the drilled hole?

A: Pilot (hole and bit)

Q19: How should flush fasteners sit inside the countersunk hole?

A: Flush or slightly above … but never below (risk b/c metal skin already too thin)

Thank You

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https://github.com/DataResearchLabs/study_guide_intro_aerospace_mfg